

Aryan Ahlawat

Bachelor of Computing (Honours) • AI Specialization • Economics Minor

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001 EDUCATION

Queen's University Bachelor of Computing (Honours), AI Specialization Stream Kingston, ON • 2024 – 2028

002 EXPERIENCE

Software Developer Co-op [Co-operators](#) May 2026 – Aug 2026
AWS Lambda • Lex • Python • Postman

- Developing RESTful Prefill API integrations to enable real-time auto-population of insurance quoting forms, orchestrating data retrieval across multiple internal services and third-party endpoints to eliminate manual entry and reduce quoting workflow completion time
- Driving spec-driven development for API documentation, building automation that generates reference docs directly from OpenAPI specifications so published documentation stays in sync with service contracts without manual upkeep

Design Team Engineer [QMIND](#) Sep 2025 – Apr 2026
LangChain • HuggingFace • RAGAS • Python

- Developed a Cognitive Retrieval-Augmented Generation (RAG) system, using LangChain agents to enable dynamic reasoning and multi-hop question answering over client data
- Engineered a hybrid retrieval pipeline combining BM25 sparse search and bi-encoder dense embeddings, optimizing context recall for multi-hop question answering tasks
- Integrated reciprocal rank fusion (RRF) and a cross-encoder re-ranking model from Hugging Face to enhance retrieval precision and accuracy

Machine Learning Co-op [Acetech](#) May 2025 – Aug 2025
PyTorch • Pandas • SQL • NumPy

- Developed Multi-Layer Perceptron (MLP) regression models in PyTorch to predict lab test durations, tuning hyperparameters to achieve a **12% reduction** in Mean Squared Error loss against the previous baseline
- Preprocessed and engineered features from an extensive SQL dataset; built automated data cleaning pipelines with Pandas, SQLAlchemy, and NumPy to impute missing values and detect outliers

Drive Control Developer [Queen's Knights Robotics](#) Oct 2024 – Mar 2025
FreeRTOS • MATLAB • C

- Programmed low-level robotic drive-control systems in C, implementing PID controllers to ensure smooth acceleration and precise velocity control
- Developed motion algorithms that improved autonomous navigation efficiency, reducing path deviations for competition tasks by almost **20%**

003 PROJECTS

Payment Optimization Engine [Hack the 6ix](#) [devpost](#) ✓ Jul 2026
Python • FastAPI • PuLP/CBC • Pydantic • Streamlit

- Built a deterministic multi-card payment router on integer-cent arithmetic, formulating a month of spend allocation as an integer program (PuLP/CBC) under hard credit-limit, utilization-ceiling, and sign-up-bonus constraints, with a greedy solver as a verified fallback
- Fine-tuned a small language model on synthetically generated data to parse natural-language financial goals into a validated weight-and-constraint schema, using the solver as a downstream verifier to score decision-match accuracy against a prompted large model

Visualizelt [QHacks](#) **WINNER • MAYOR'S INNOVATION AWARD** [devpost](#) ✓ Feb 2025
YOLOv8 • PyTorch • Stable Diffusion • OpenCV

- Engineered a real-time computer vision pipeline using YOLOv8 (Nano) for millisecond-latency object detection, serving as the initialization anchor for robust OpenCV CSRT tracking
- Integrated Stable Diffusion inpainting to generate context-aware texture overlays, dynamically mapping stylized graphics onto detected regions with high geometric consistency

004 SKILLS & CERTIFICATIONS

Languages	Python, Java, C, SQL, JavaScript, PyTorch, LangChain, FastAPI, Flutter, Scikit-Learn
Tools • Platforms	AWS, Azure, Oracle Cloud, Git, Postman, OpenAPI, MATLAB, OpenCV, Databricks, Docker
Certifications	Oracle Cloud Infrastructure Foundations, Microsoft AZ-900, NVIDIA Building RAG Agents with LLMs